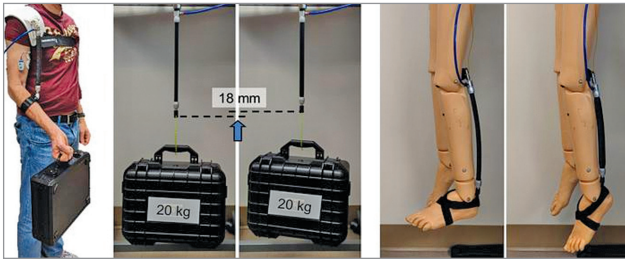


Implantable microphone for fully internal cochlear implants

A collaborative effort by MIT, Massachusetts Eye and Ear, Harvard Medical School, and Columbia University has resulted in the development of an implantable microphone, a crucial step towards fully internal cochlear implants. Named UmboMic, the sensor is crafted from biocompatible piezoelectric material, detecting minute movements on the eardrum's underside. Enhanced by a low-noise amplifier, it targets the umbo in the middle ear, enabling precise detection of nanometre-scale vibrations. Successful tests on human cadaver ear bones validate its performance within speech ranges, showcasing its potential for broader adoption.

Fluidic engine for soft robots for assistive devices

Researchers at North Carolina State University have developed a lightweight fluidic engine for muscle-mimicking soft robots used in assistive devices. Unlike traditional engines,



Muscle-mimicking soft robots (Credit: <https://solondais.fr>)

this innovative design generates substantial force independently, up to 580 Newtons, without external power sources. Operating on hydraulic principles, the engine pumps oil into a chamber within the robot, replicating muscle movements. Powered by a high-torque motor, it efficiently converts electrical power into fluidic force. Their proof-of-concept tests demonstrated unprecedented force output and efficiency, making it ideal for enhancing limb mobility.

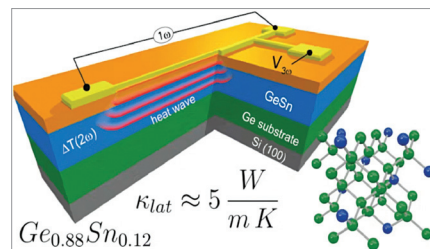
Greener, cheaper lithium-ion batteries

Researchers at Dalhousie University have developed a greener and more cost-effective manufacturing process for lithium-ion cathode materials. Traditionally, producing these materials consumes substantial energy and water, contributing to environmental impacts. Seeking sustainable alternatives, the team utilised the Canadian light source (CLS) to explore an all-dry process. By heating dry materials without water, they achieved commercial-grade cathode materials while conserving energy and minimising waste. Despite challenges in achieving homogeneity compared to wet processes, the team

optimised their method using various oxides and heating conditions, observed through CLS's Brockhouse beamline. Partnering with NOVONIX, they established a pilot-scale facility in Dartmouth, projecting significant cost reductions and environmental benefits, advancing battery technology towards lower-cost, eco-friendly solutions.

New materials convert electronic waste heat into energy

The rising demand for electronic devices results in increased energy consumption, primarily released as low-temperature heat below 80°C. Overcoming the inefficiencies and technological barriers in harnessing this waste heat for electricity remains challenging. A collaborative effort involving Forschungszentrum Jülich, IHP—



Researchers have come up with a new alloy which could, they say, lead to computer chips that can harvest their own waste heat energy (Credit: Concepcion et al)

Leibniz Institute for High-Performance Microelectronics, and universities in Germany, Italy, and the UK has made significant strides in developing thermoelectric materials compatible with CMOS chip production. These materials convert temperature differences directly into electrical energy, recycling waste heat from electronics. GeSn alloys, with their low thermal conductivity, enhance thermoelectric performance by maintaining a substantial temperature gradient, aligning with the goal of integrating photonics, electronics, and thermoelectrics on silicon-based chips for enhanced sustainability.

EM wave propagation to boost 5G/6G networks

Researchers from the Smart and Wireless Applications and Technologies Group (SWAT-UGR) at Aalborg University delve into electromagnetic wave propagation for advancing 5G and 6G networks and Industry 4.0 automation in factories. Their studies, published in IEEE journals, focus on estimating signal parameters using toroidal arrays and analysing millimetre wave behaviour in industrial settings with obstacles. These insights are pivotal as networks face saturation risks due to escalating user and device numbers. Proposed solutions involve deploying services in higher-frequency bands to enhance speed and stability, enabling innovations like robotic surgery and virtual reality.